

AERIS Operator Cheat Sheet

Printable quick reference for CLI, scripts, and the new section-count study

Use most often: aeris geometry generate, aeris dataset generate, aeris dataset inspect

Use occasionally: aeris version, aeris pipeline smoke, visualization scripts, integrity checks

A. Core CLI — Day-to-Day

Use case	Command
Check install/version	aeris version
Top-level help	aeris --help
Geometry help	aeris geometry --help
Dataset help	aeris dataset --help
Pipeline help	aeris pipeline --help
Generate one baseline geometry	aeris geometry generate -c configs/geometry/baseline_bwb.yaml
Generate one sampled geometry	aeris geometry generate -c configs/geometry/wing_bwb.yaml
Small LHS dataset	aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 20 --sampler lhs_v1 --sampler-seed 123
Fast production dataset (no plots)	aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 500 --sampler lhs_v1 --sampler-seed 123 --no-save-plot --name stress_wing_bwb_n500_lhs123_noplots
AeroSandbox-enabled batch	aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 20 --sampler lhs_v1 --sampler-seed 123 --no-save-plot --build-aerosandbox --name stress_wing_bwb_n20_lhs123_asb
Random sampler test	aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 20 --sampler random_v1 --sampler-seed 123 --no-save-plot --name random_test_n20
Inspect dataset	aeris dataset inspect --dataset data/datasets/stress_wing_bwb_n500_lhs123_noplots
Inspect ASB-enabled dataset	aeris dataset inspect --dataset data/datasets/stress_wing_bwb_n20_lhs123_asb
Smoke test (good config)	aeris pipeline smoke -c configs/geometry/baseline_bwb.yaml
Smoke test (bad config)	aeris pipeline smoke -c configs/smoke/bad.yaml

B. Useful Scripts — Diagnostics / Visualization / Checks

Use case	Command
Visualize baseline airplane	python scripts/bwb_segmented_v1/visualize_airplane.py --mode baseline
Visualize one dataset airplane	python scripts/bwb_segmented_v1/visualize_airplane.py --mode dataset --dataset data/datasets/stress_wing_bwb_n500_lhs123_asb --geometry-id geom_00001
Visualize 5 random airplanes sequentially	DATASET="data/datasets/stress_wing_bwb_n500_lhs123_asb" ls "\$DATASET/geometry" shuf head -n 5 while read GEOM_ID; do echo "Visualizing \$GEOM_ID" python scripts/bwb_segmented_v1/visualize_airplane.py \ --mode dataset --dataset "\$DATASET" --geometry-

	id "\$GEOM_ID" done
Bad-config failure check	python scripts/generic/check_bad_config_failure.py
Architecture integrity check	python scripts/generic/check_architecture_integrity.py
Metadata regeneration check	python scripts/generic/check_metadata_regeneration.py
Seed variation check	python scripts/generic/check_seed_variation.py
Large dataset sanity check	python scripts/generic/check_large_dataset_sanity.py
Audit a dataset	python scripts/generic/audit_dataset.py
Run one case and save debug artifacts/plot	python scripts/bwb_segmented_v1/run_single_case_and_plot_3 d.py

C. New Section-Count Study Scripts

Use these when choosing how many sections / n_points to use for production geometry generation.

Run the parametric study:

```
python scripts/section_study.py \
  --config configs/geometry/baseline_bwb.yaml \
  --sections 15 25 40 60 100 \
  --n 1 \
  --sampler lhs_v1 \
  --sampler-seed 123 \
  --name baseline_section_study \
  --build-aerosandbox \
  --force-clean
```

Pick the lowest converged default:

```
python scripts/section_study_pick_default.py \
  --summary-csv data/debug/section_study/baseline_section_study/section_study_summary.csv \
  --require-success
```

D. Very Useful Shell Helpers

```
latest_dataset="$(ls -td data/datasets/* | head -n 1)"
echo "$latest_dataset"
find "$latest_dataset/geometry" -maxdepth 1 -type d | wc -l
wc -l "$latest_dataset/metadata.csv"
wc -l "$latest_dataset/failures.csv"
tail -n 20 "$latest_dataset/logs/dataset.log"
```

E. Practical Notes

- Prefer `--sampler lhs_v1 --sampler-seed 123` over old `lhs_seed` style.
- Use `--no-save-plot` for real batch runs. Plotting is a major runtime tax.
- Use geometry for one airplane, dataset for many airplanes, pipeline for sanity/workflow checks.
- Dataset visualization should use the dataset's own config snapshot + metadata sample.
- For section-count studies, think in three levels: debug (15–25), standard production (35–40), high-resolution reference (80–100).

Generated for print-friendly use. Update this sheet whenever CLI or script behavior changes.